AI-Assisted Patient Intake and Pre-Consultation Summaries

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Problem

In most hospitals, patients describe their condition on paper or digital forms, often several times per visit, before a doctor sees them. Staff transcribe and sort them by hand, and doctors read them and the patient's history separately. Under staff shortages details get lost or arrive late, and in long-term care nurses spend up to a third of their working time on documentation (Schwabe et al., 2026). Patients, in turn, have to describe symptoms in medical terms they often do not understand. EHR systems store data but do not summarise it, and existing intake tools (e.g. Ada, Infermedica) send data to one vendor model with no control over where it is processed.

Idea

An LLM-based intake assistant that talks to the patient in plain language, combines the answers with their history from the hospital database, and hands the doctor a structured summary (complaint, history, warnings such as allergies). The architectural core is a **privacy-aware LLM gateway**: a **pseudonymiser** strips identifiers, a **router** sends identifiable data to a small **local model** and de-identified text to a stronger **cloud model**, and **guardrails** check both directions. Around it: a **patient chat client**, an **intake orchestrator** driving the dialogue, a read-only **EHR adapter** (HL7 FHIR), and a **summary service with clinician dashboard**. Style: layered modular monolith, so all privacy rules live in one gateway. Key trade-off: **privacy vs. summary quality vs. cost**, made explicit as routing rules. Quality attributes: privacy (Art. 9 GDPR), correctness, usability for non-experts. Out of scope: diagnosis or treatment advice; writing back to the hospital database.

Benefits

- **Less documentation work for staff** – addresses manual transcription and sorting of intake data; shown by time per intake, manual form vs. assistant; a speech assistant cut nursing documentation time by about 28% (Schwabe et al., 2026).
- **Faster, better-informed consultations** – addresses scattered information; demonstrated by accuracy of summaries against the synthetic ground-truth records.
- **Easier intake for patients** – addresses medical jargon on forms; shown by test-user feedback.

Context: Europe's digital health market grows a projected 22.3% p.a. (2024–2030), partly driven by staff shortages (Grand View Research).

Action

1. W1–3: requirements from interviews and surveys (clinicians or medical students as proxy users); quality-attribute scenarios; architecture and ADRs.
2. W4–8, **minimum deliverable:** chat client, orchestrator, gateway with routing, and summary view on synthetic patient data.
3. W9–10: pseudonymiser, guardrails, EHR adapter.
4. W11–12: evaluation against the benefits; W13: buffer.

Stack: Python/FastAPI, Ollama (local model), OpenRouter (cloud model), HAPI FHIR, Synthea synthetic patients, React, Docker. No real patient data is used.

AI usage statement. Claude Opus 5.5 (via claude.ai) for brainstorming, restructuring the text into this template, and checking cited sources. All content was reviewed and verified by the team.